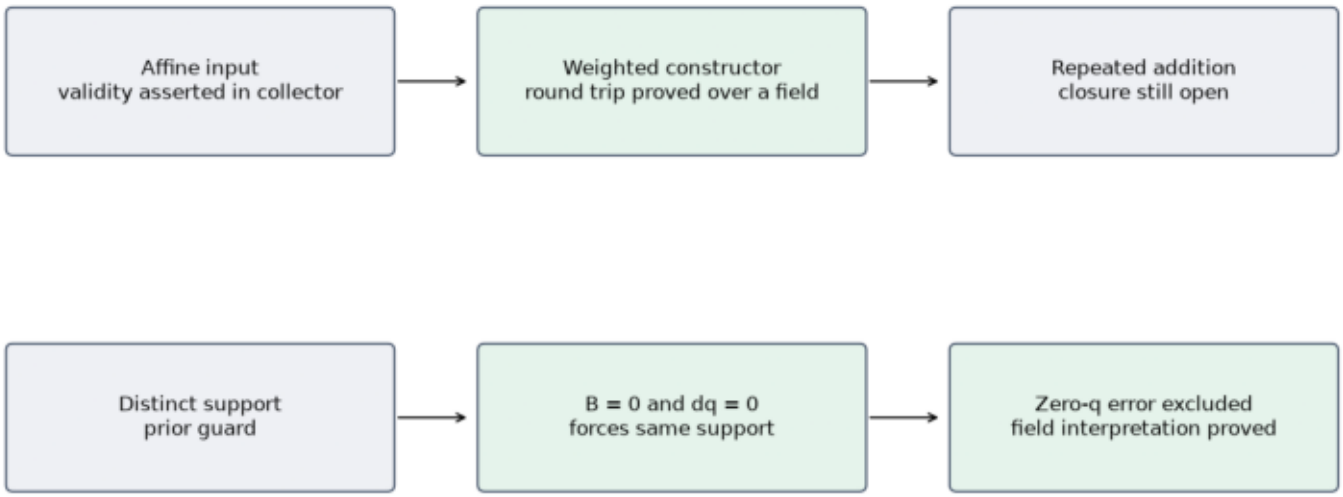
Constructor invariants and branch exclusion

Five field-level declarations compiled; separate axiom readback exited 0. Only propext, Classical.choice and Quot.sound were reported. No scientific trial or benchmark was run. Independent review is pending.

Results: weighted-coordinate round trip; preservation of Mumford divisibility under reweighting; preservation under negation; B=0 and dq=0 force equal support; hence the shared_zero_q_delta guard is excluded after the different-support test.

These propositions interpret the exact source expressions over a field. They do not establish UInt64 correspondence or closure of every collector transition.



Green: new producer-checked field lemmas. Machine correspondence and independent review remain open.

Evidence: InvariantBranches.lean; receipt.json; readback.json; sources.json.
First failed compile retained under attempt1/. Source mapping: branches.json.
DOI: 10.1007/s13389-024-00351-7. Full derivations and obligations: REPORT.md.

Invariant boundary and next proof obligations

Entry: collector lines 153–160 assert affine validity before conversion. Jacobian::valid checks degree, monicity and polynomial remainder; it does not check primality, canonical words or smoothness. Degree-two weighting/decoding now has a field-level preservation lemma.

Transitions: identity returns its operand. Other dispatcher branches feed scalar multiplication, jump construction, initialization, restart and walk steps. A single closed dispatcher invariant would support induction over these loops. That closure remains unproved.

1+1: common-x ordinate alternatives and doubling tangent validity need a source bridge. Generic 1+2: nonzero output weights factor from K and nonzero input weights; numerator congruence is the next local target.

Same support: inverse and equal-v precede mixed. The prior weighted same-support theorem handles zero slope; the present result handles the different-support shared-zero-q branch. They are distinct exclusions.

Shared positive support: extracted-point validity, matching ordinates and both secondary additions remain obligations. Generic and degree-drop 2+2 weight factors are nonzero under their guards; output congruence and divisor-class meaning remain separate.

Scope: one failed compiler attempt is retained; its sorryAx output is rejected. A source comparison initially found drift in an older three_leads.rs; the current worktree version was then pinned instead. No unsupported historical source identity is assumed.

Next: independent proposition/source review, then machine-operation correspondence and generic 1+2 output congruence. Report no cost ratio from proof-build timings.